

Distributed Microservices-Based Restaurant Website

ABSTRACT

OBJECTIVE:-

The main objective of the Distributed Microservices-Based Restaurant Website is to provide a simple and efficient online platform where customers can view food items, place orders, make online payments, and track their orders. To provide an online food ordering system for customers. To display the restaurant's food menu, prices, descriptions, and availability. To allow customers to add food items to a shopping cart. To allow customers to place food orders online. To provide online payment options, including card and other supported digital payments. To allow customers to view order status and order history. Overall my project is to Reduce manual ordering processes and provide a faster and more convenient restaurant ordering experience.

EXISTING SYSTEM:-

The existing restaurant ordering system is mainly based on traditional manual processes or a centralized online system. Customers may need to visit the restaurant, call the restaurant, or use a basic ordering platform to place their orders. Restaurant staff manually manage menus, customer details, orders, payments, and delivery information. If Any Hike Of Prices also we will not have any update if we want to know about that we should travel. Any updates can't be know for

any one fast. And user may or maynot found our restaurant so so he must visit restaurant physically.

DRAWBACKS:-

- Manual Order Processing – Orders may need to be entered manually, increasing staff workload.
- Human Errors – Mistakes can occur in food selection, quantity, pricing, and customer details.
- Payment Limitations – Traditional systems may provide limited online payment options.
- Difficult Data Management – Managing customer, food, and order records manually can become difficult as data increases.
- Maintenance Problems – Changes or updates to one part of a tightly connected system can affect other parts.
- Limited Scalability – A centralized system may face difficulties when the number of users and orders grows.

PROPOSED SYSTEM:-

The proposed system is a Distributed Microservices-Based Restaurant Website designed to provide a convenient and efficient online food ordering platform. The website allows customers to register and log in, view the restaurant menu, check food prices and availability, add items to a cart, place orders, and make secure online payments using supported methods such as debit cards, credit cards, and UPI through a payment gateway.

Customers can also track their order status from order placement to delivery and view their previous orders. The

system uses a microservices architecture, where user management, menu management, order processing, payment processing, and notifications are handled as separate services.

The proposed system reduces manual work, minimizes errors, improves order processing, and provides a better overall restaurant ordering experience for customers and administrators.

ROLES & RESPONSIBILITIES:

The proposed restaurant website contains different users and services. Each role has specific responsibilities to make the online food ordering process simple and efficient.

1. Customer

Role: The customer is the main user of the website who orders food online.

Responsibilities:

- Register and create an account.
- Login securely to the website.
- View the restaurant menu.
- Search and select food items.
- Check food prices and availability.
- Add food items to the shopping cart.
- Modify food quantity or remove items.
- Place food orders.
- Enter delivery details.
- Make online payments using supported payment methods.

- Check payment status.
- Track the order status.
- View previous orders.

2. Restaurant Admin

Role: The admin manages the overall restaurant website and its operations.

Responsibilities:

- Manage customer accounts.
- Add new food items.
- Update food names, prices, descriptions, and images.
- Remove unavailable food items.
- Manage food categories.
- Check incoming orders.
- Accept and process orders.
- Update order status.
- Check payment status.
- Manage restaurant information.

3. Delivery Staff

Role: Delivery staff are responsible for delivering prepared food to customers.

Responsibilities:

- View assigned orders.
- Collect prepared food from the restaurant.
- Check order details.

- Deliver food to the customer's address.
- Update delivery status.
- Mark the order as delivered.

4. User Service

Role: Manages customer-related information and authentication.

Responsibilities:

- Handle user registration.
- Manage login and logout.
- Authenticate users.
- Manage customer profiles.
- Maintain user information.
- Provide user-related REST APIs.

5. Menu Service

Role: Manages all restaurant food and menu information.

Responsibilities:

- Add food items.
- Update food details.
- Delete food items.
- Manage categories.
- Manage food prices.
- Check food availability.
- Display menu information through REST APIs.

6. Order Service

Role: Manages the complete food-ordering process.

Responsibilities:

- Create new orders.
- Manage cart/order details.
- Calculate order totals.
- Store order information.
- Update order status.
- Maintain order history.
- Provide order-related REST APIs.

7. Payment Service

Role: Handles online payment processing.

Responsibilities:

- Receive payment requests.
- Connect with the payment gateway.
- Support supported payment methods such as cards and UPI.
- Verify payment results.
- Maintain payment transaction status.
- Send payment status to the Order Service.

8. Notification Service

Role: Provides important updates to customers.

Responsibilities:

- Send order confirmation.
- Send payment confirmation.
- Notify customers about order status.
- Send delivery updates.

- Provide important system notifications.